Chapter 9 — Acid-Base Chemistry

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 204–206

Q.1 — Multiple Choice Questions (with correct option and reason)

I. For $NH_3(g) + H_2O(l) \rightarrow NH_4^+(aq) + OH^-(aq)$:

(a) NH_3 is the acid, H_2O is the base.
(b) NH_3 is the base, H_2O is the acid.
(c) NH_4^+ is the base, OH^- is the acid.
(d) H_2O is the base, OH^- is the acid.

Answer: (b) NH_3 is the base, H_2O is the acid. — NH_3 accepts a proton (base) and H_2O donates a proton (acid).

II. The pH of 10^{-3} mol dm^{-3} aqueous H_2SO_4 is:

(a) 3.0
(b) 2.7
(c) 2.0
(d) 1.5

Answer: (b) 2.7 — Assuming complete ionization, $[H^+] = 2 \times 10^{-3}$ M; pH = $-\log(2 \times 10^{-3}) = 2.7$.

III. Ksp of AgCl is 2.0×10^{-10} mol^2 dm^{-6}. The maximum $[Ag^+]$ in the solution is:

(a) 2.0×10^{-10} mol dm^3
(b) 1.41×10^{-5} mol dm^{-3}
(c) 1.0×10^{-10} mol dm^{-3}
(d) 4.0×10^{-20} mol dm^{-3}

Answer: (b) 1.41×10^{-5} mol dm^{-3} — For AgCl, Ksp = s^2, so $s = \sqrt{(2.0 \times 10^{-10})} = 1.41 \times 10^{-5}$ mol dm^{-3}.

IV. Which indicator is typically used for titrations involving strong acids and strong bases?

(a) Methyl red
(b) Phenolphthalein
(c) Bromothymol blue
(d) Litmus solution

Answer: (b) Phenolphthalein — Phenolphthalein is commonly used because the steep pH change at equivalence covers its colour-change range.

Bromothymol blue (c) is also suitable for strong acid–strong base titrations; the question has more than one acceptable indicator.

V. Which of the following is the conjugate base of water?

(a) $OH^-(aq)$
(b) $H^+(aq)$
(c) $H_2O(l)$
(d) $H_3O^+(aq)$

Answer: (a) $OH^-(aq)$ — Water loses one H^+ to form its conjugate base, OH^-.

VI. Which of the following is a Lewis acid but not a Bronsted-Lowry acid?

(a) HCl
(b) NH_3
(c) $AlCl_3$
(d) H_2O

Answer: (c) $AlCl_3$ — $AlCl_3$ accepts an electron pair (Lewis acid) but has no proton to donate.

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VII. In an acid-base titration, the equivalence point is reached when:

(a) pH of the solution is 7.0.
(b) The indicator changes colour.
(c) Equal volumes of acid and base have been added.
(d) The reaction stops.

Answer: (b) The indicator changes colour. — In practice the indicator colour change (end point) is taken to signal the equivalence point.

Strictly, the equivalence point is the stoichiometric point; the indicator colour change is the end point. The question conflates the two.

VIII. If the concentration of Cl^- ion in a solution is increased, the solubility of AgCl will:

(a) Decrease
(b) Increase
(c) Remain unchanged
(d) Become zero

Answer: (a) Decrease — The common ion Cl^- shifts $AgCl(s) \rightleftharpoons Ag^+ + Cl^-$ to the left, decreasing solubility.

IX. Which pair can act as a conjugate acid-base pair according to Bronsted-Lowry theory?

(a) HCl and NaOH
(b) NH_3 and NH^+
(c) H_2O and H_2SO_4
(d) H_2O and CH_4

Answer: (b) NH_3 and NH^+ — NH_3 and NH^+ differ by one proton, so they are a conjugate acid-base pair.

X. If the pH of a solution is 11, what is the $[OH^-]$ concentration?

(a) 1×10^{-3} M
(b) 1×10^{-11} M
(c) 1×10^{-2} M
(d) 1×10^{-14} M

Answer: (a) 1×10^{-3} M — $pOH = 14 - 11 = 3$, so $[OH^-] = 10^{-3}$ M.

XI. Which of the following pairs forms a buffer solution?

(a) HCl and NaCl
(b) CH_3COONa and CH_3COOH
(c) NaOH and HCl
(d) NH_3 and Na_2SO_4

Answer: (b) CH_3COONa and CH_3COOH — A weak acid (CH_3COOH) and its salt (CH_3COONa) form an acidic buffer.

Q.2 — Short Answer Questions

a. Define with an example: (i) Ionization constant (ii) Solubility product (iii) Common ion effect (iv) Acid-base indicator.

(i) Ionization constant (Ka or Kb): the equilibrium constant for ionization of a weak acid/base; e.g. $K_a = [H^+][A^-]/[HA]$ for CH_3COOH.

(ii) Solubility product (Ksp): the product of ion concentrations in a saturated solution of a sparingly soluble salt, each raised to its coefficient; e.g. $K_{sp}(AgCl) = [Ag^+][Cl]$.

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(iii) Common ion effect: suppression of ionization of a weak electrolyte by adding a strong electrolyte with a common ion; e.g. adding CH_3COONa suppresses CH_3COOH ionization.

(iv) Acid-base indicator: a weak organic acid/base whose unionized and ionized forms have different colours; e.g. phenolphthalein.

The book prints two items as “iii)”; both the common ion effect and the indicator are answered.

b. Differentiate between: (i) Hydrolysis and dissolution (ii) Acidic and basic buffer solutions.

(i) Dissolution is the physical dispersal of a solute in a solvent; hydrolysis is a chemical reaction of salt ions with water that changes pH (e.g. $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$).

(ii) An acidic buffer contains a weak acid and its salt (pH < 7, e.g. CH_3COOH/CH_3COONa); a basic buffer contains a weak base and its salt (pH > 7, e.g. NH_3/NH_4Cl).

c. Explain the concept of conjugate acid-base pairs. How are they related in terms of proton transfer?

A conjugate acid-base pair consists of two species that differ by one proton. When an acid donates H^+ it becomes its conjugate base; when a base accepts H^+ it becomes its conjugate acid. Example: HA/A^- and NH_4^+/NH_3.

d. What is the relationship between the strength of an acid and the strength of its conjugate base?

The stronger the acid, the weaker its conjugate base, and vice versa. A strong acid ionizes completely, so its conjugate base has little tendency to accept a proton.

e. For the reactions, identify Arrhenius, Bronsted-Lowry and/or Lewis bases: (i) $NaOH(s) \rightarrow Na^+ + OH^-$ (ii) $HF + H_2O \rightarrow F^- + H_3O^+$ (iii) $H^+ + NH_3 \rightarrow NH_4^+$.

(i) NaOH is an Arrhenius base (produces OH^- in water); OH^- is also a Bronsted-Lowry and Lewis base.

(ii) H_2O acts as a Bronsted-Lowry base (accepts H^+) and a Lewis base (donates a lone pair).

(iii) NH_3 is a Bronsted-Lowry base (accepts H^+) and a Lewis base (donates its lone pair).

f. Identify whether water behaves as an acid or a base in: (i) $H_2O + HCl \rightarrow H_3O^+ + Cl^-$ (ii) $NH_3 + H_2O \rightarrow NH_4^+ + OH^-$ (iii) $HNO_3 + H_2O \rightarrow H_3O^+ + NO_3^-$ (iv) $CH_3COOH + H_2O \rightarrow CH_3COO^- + H_3O^+$.

(i) Base (accepts H^+). (ii) Acid (donates H^+ to NH). (iii) Base (accepts H^+). (iv) Base (accepts H^+). Water is amphoteric.

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g. Which salt would dissolve more readily in acidic solution: barium carbonate or copper sulfide? Explain. Ksp($BaCO_3$) = 1.1 × 10^{-10}, Ksp(CuS) = 8 × 10^{-34}.

Barium carbonate dissolves more readily in acid. The carbonate ion reacts with H^+ ($CO_3^{2-} + 2H^+ \rightarrow CO_2 + H_2O$), removing CO_3^{2-} and shifting dissolution forward. CuS has an extremely small Ksp and is not appreciably dissolved by non-oxidising acids.

h. Why does the common ion effect decrease the solubility of a less soluble salt?

Adding a common ion increases the concentration of one of the ions in the dissolution equilibrium. By Le Chatelier's principle the equilibrium shifts toward the solid, so less salt dissolves.

i. State the basic principle of solubility product. Mention factors affecting solubility product.

In a saturated solution of a sparingly soluble salt at a given temperature, the product of the ion concentrations (each raised to its stoichiometric coefficient) is constant (Ksp). Ksp depends on temperature and the nature of the salt/solvent; it is not changed by adding a common ion (though solubility changes).

j. Prove by equations what happens when Na_2CrO_4 is added to a saturated solution of $PbCrO_4$.

$PbCrO_4(s) \rightleftharpoons Pb^{2+}(aq) + CrO_4^{2-}(aq)$. Adding Na_2CrO_4 increases $[CrO_4^{2-}]$ (common ion). The ionic product exceeds Ksp, so the equilibrium shifts left and more $PbCrO_4$ precipitates: $Pb^{2+} + CrO_4^{2-} \rightarrow PbCrO_4(s)$.

k. According to the Lewis concept, can BF_3 act as an acid? Is this statement correct?

Yes. Boron in BF_3 has only six electrons and an empty orbital, so it can accept an electron pair from a Lewis base (e.g. NH_3), forming $F_3B{\leftarrow}NH_3$. Hence BF_3 is a Lewis acid.

l. If $[H^+] = 1 \times 10^{-5}$ M, what is the pH?

$pH = -\log[H^+] = -\log(10^{-5}) = 5$.

Descriptive / Long Questions

Q.3 Describe the Bronsted-Lowry theory of acids and bases. Provide examples of conjugate acid-base pairs and explain their relationship.

According to Bronsted-Lowry, an acid is a proton (H^+) donor and a base is a proton acceptor. Acid-base reactions involve transfer of a proton from the acid to the base.

Example: $HCl + H_2O \rightarrow H_3O^+ + Cl^-$. HCl donates H^+ (acid) and H_2O accepts it (base). The pairs are HCl/Cl^- and H_3O^+/H_2O.

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Conjugate pairs differ by one proton: the acid forms its conjugate base after losing H^+, and the base forms its conjugate acid after gaining H^+. A strong acid has a weak conjugate base.

Q.4 Define the Lewis theory of acids and bases. How does it differ from Bronsted-Lowry? Give examples of Lewis acids and bases that do not involve proton transfer.

A Lewis acid is an electron-pair acceptor and a Lewis base is an electron-pair donor. The theory is broader than Bronsted-Lowry because it includes reactions without proton transfer.

Example: $BF_3 + NH_3 \rightarrow F_3B{\leftarrow}NH_3$. BF_3 accepts a lone pair (Lewis acid); NH_3 donates it (Lewis base). No proton is transferred. Other Lewis acids: $AlCl_3$, Fe^{3+}; Lewis bases: NH_3, H_2O, OH^-.

Q.5 Discuss applications and implications of the common ion effect in various fields.

(i) Qualitative analysis: precipitation of group II sulfides is controlled by adding HCl (common H^+ suppresses H_2S ionization). (ii) Buffers: a weak acid and its salt resist pH change via the common ion effect. (iii) Purification of NaCl: passing HCl gas into saturated brine precipitates pure NaCl (common Cl^-). (iv) Soap manufacture: adding NaCl precipitates soap (salting out). (v) Solubility control: adding a common ion decreases the solubility of a sparingly soluble salt.

Q.6 What is the solubility product for sparingly soluble salts? Give its two applications.

For a sparingly soluble salt $A_xB_y(s) \rightleftharpoons xA^{n+}(aq) + yB^{m-}(aq)$, the solubility product is $Ksp = [A^{n+}]^x[B^{m-}]^y$ at a given temperature. It is constant for a saturated solution.

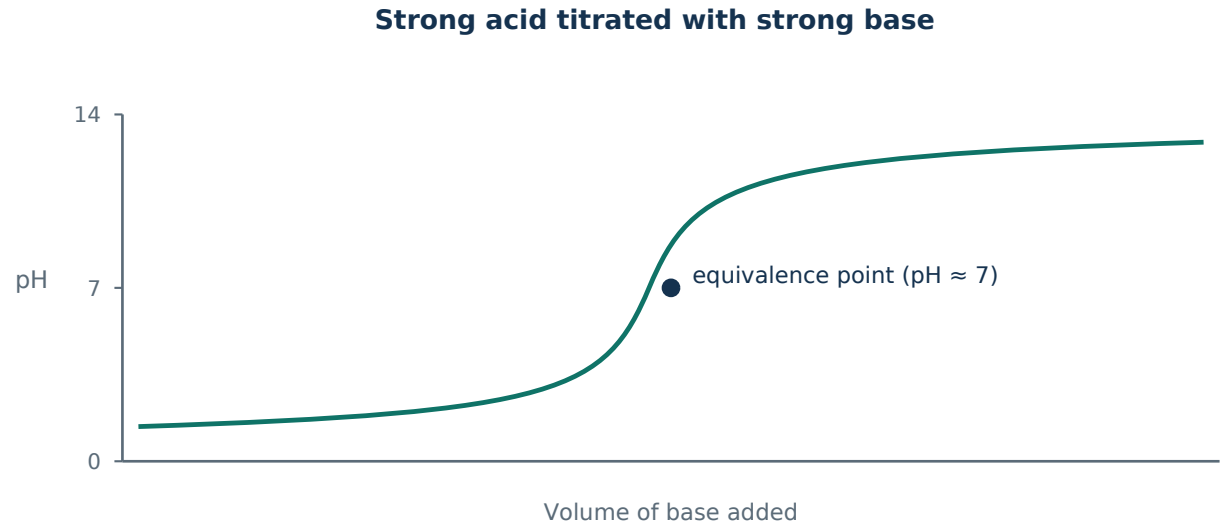
Applications: (1) Predicting precipitation: if the ionic product > Ksp, precipitation occurs; if < Ksp, no precipitate. (2) Calculating solubility and controlling separation of ions in qualitative analysis by adjusting the common-ion concentration.

Q.7 Describe the general shape of a titration curve for a strong acid titrated with a strong base. How can you identify the equivalence point?

The curve starts at low pH (excess strong acid). As base is added the pH rises slowly, then very steeply near the equivalence point, and finally levels off at high pH (excess base). For a strong acid–strong base titration the equivalence point is at pH ≈ 7.

The equivalence point is identified at the midpoint of the steep vertical portion of the curve (the inflection point). An indicator whose colour-change range lies within this steep region (e.g. phenolphthalein or bromothymol blue) marks the end point.

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Numerical Problems (fully worked)

Q.8 A buffer has pH 5.0 and is made from a weak acid HA with pKa 4.8. What is the ratio [A⁻]/[HA]?

Given: pH = 5.0; pKa = 4.8

Formula: Henderson–Hasselbalch: pH = pKa + log([A⁻]/[HA])

log([A⁻]/[HA]) = pH − pKa = 5.0 − 4.8 = 0.2.

[A⁻]/[HA] = 10^{0}.2 = 1.58.

Answer: [A⁻]/[HA] ≈ 1.6

Q.9 Calculate the solubility of PbI_2 in water. Ksp = 1.4×10^{-8}.

Given: Ksp(PbI_2) = 1.4×10^{-8}

Formula: $PbI_2(s) \rightleftharpoons Pb^{2+} + 2I^-$; Ksp = $[Pb^{2+}][I^-]^2 = 4s^3$

Let solubility = s. Then $[Pb^{2+}] = s$, $[I^-] = 2s$; Ksp = $s(2s)^2 = 4s^3$.

$s^3 = 1.4 \times 10^{-8}/4 = 3.5 \times 10^{-9} \rightarrow s = (3.5 \times 10^{-9})\text{^}(1/3) = 1.52 \times 10^{-3}$ mol dm^{-3}.

Answer: Solubility ≈ 1.5×10^{-3} mol dm^{-3}

Q.10 The molar solubility of Ag_2CrO_4 in pure water at 298 K is 6.5×10^{-5} mol dm^{-3}. Calculate Ksp.

Given: s = 6.5×10^{-5} mol dm^{-3}

Formula: $Ag_2CrO_4(s) \rightleftharpoons 2Ag^+ + CrO_4^{2-}$; Ksp = $[Ag^+]^2[CrO_4^{2-}] = 4s^3$

$[Ag^+] = 2s$, $[CrO_4^{2-}] = s$; Ksp = $(2s)^2(s) = 4s^3$.

Ksp = $4 \times (6.5 \times 10^{-5})^3 = 4 \times 2.746 \times 10^{-13} = 1.10 \times 10^{-12}$.

Answer: Ksp ≈ 1.1×10^{-12} mol^3 dm^{-9}

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Quick Check Solutions (in-chapter practice boxes)

Quick Check 9.1 (book p. 188)

a. Identify conjugate acid-base pairs in: (i) $HCN + H_2O \rightarrow H_3O^+ + CN^-$ (ii) $CH_3COOH + H_2O \rightarrow H_3O^+ + CH_3COO^-$ (iii) $H_2S + H_2O \rightarrow H_3O^+ + HS^-$ (iv) $HCl + HCO_3^- \rightarrow H_2CO_3 + Cl^-$.

(i) HCN/CN^- and H_3O^+/H_2O. (ii) CH_3COOH/CH_3COO^- and H_3O^+/H_2O. (iii) H_2S/HS^- and H_3O^+/H_2O. (iv) HCl/Cl^- and H_2CO_3/HCO_3^-.

b. Identify Lewis acid/base and write equations: (i) $SO_3 + O^{2-}$ (ii) $HCl + H_2O$ (iii) $BF_3 + NH$.

(i) SO_3 (acid) + O^{2-} (base) → SO_4^{2-}. (ii) $HCl + H_2O \rightarrow H_3O^+ + Cl^-$; H_2O donates a lone pair (Lewis base), H^+ accepts (Lewis acid). (iii) BF_3 (acid) + NH_3 (base) → $F_3B{\leftarrow}NH_3$.

Quick Check 9.2 (book p. 191)

a. Equal volumes of pH 4.0 and pH 10.0 solutions are mixed. Calculate Kw at 25 °C.

Kw is a constant at a given temperature; at 25 °C it remains 1.0×10^{-14} $mol^2\ dm^{-6}$ regardless of mixing.

b. $K_w = 1.0\times10^{-14}$; $[OH^-] = 2.5\times10^{-8}$ M. Calculate $[H^+]$ and pH.

$[H^+] = K_w/[OH^-] = 1.0\times10^{-14}/2.5\times10^{-8} = 4.0\times10^{-7}$ M. $pH = -\log(4.0\times10^{-7}) = 6.40$.

c. 0.30 M HNO_3: calculate $[H^+]$, pH, $[OH^-]$, pOH at 25 °C.

$[H^+] = 0.30$ M; $pH = -\log(0.30) = 0.52$. $[OH^-] = 1.0\times10^{-14}/0.30 = 3.3\times10^{-14}$ M; $pOH = 14 - 0.52 = 13.48$.

Quick Check 9.3 (book p. 193)

a. 0.10 M formic acid has pH 2.38. Calculate Ka.

$[H^+] = 10^{-2.38} = 4.17\times10^{-3}$ M. $K_a = [H^+]^2/[HA] \approx (4.17\times10^{-3})^2/0.10 = 1.7\times10^{-4}$.

b. $[H_3O^+]$ in 0.1 M HNO_2, $K_a = 4\times10^{-4}$.

$[H_3O^+] \approx \sqrt{(K_a \cdot c)} = \sqrt{(4\times10^{-4} \times 0.1)} = \sqrt{(4\times10^{-5})} = 6.3\times10^{-3}$ M.

c. Vinegar 0.837 M CH_3COOH, $[H_3O^+] = 3.86\times10^{-3}$ M. Calculate Ka.

$K_a = (3.86\times10^{-3})^2/0.837 = 1.78\times10^{-5}$.

Quick Check 9.4 (book p. 196)

a. Impact of common ion effect on solubility.

Adding a common ion shifts the dissolution equilibrium toward the solid, decreasing the solubility of a sparingly soluble salt.

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b. NaCl is added to saturated AgCl: (i) effect on [Ag^+] (ii) explain by common ion effect.

(i) [Ag^+] decreases. (ii) Added Cl^- shifts $AgCl(s) \rightleftharpoons Ag^+ + Cl^-$ to the left, precipitating AgCl and lowering [Ag^+].

c. How a buffer maintains pH stability.

A buffer contains a weak acid and its conjugate base (or weak base and conjugate acid). Added H^+ is consumed by the base component and added OH^- by the acid component, so pH changes only slightly.

d. pH of 0.50 M HF / 0.45 M F^- buffer before and after adding 0.40 g NaOH to 1.0 dm^3. Ka(HF) = 6.8×10^{-4}.

pKa = 3.17. Before: pH = 3.17 + log(0.45/0.50) = 3.12. NaOH added = 0.40/40 = 0.010 mol; it converts HF to F^-: [HF] = 0.49, [F^-] = 0.46. After: pH = 3.17 + log(0.46/0.49) = 3.14.

e. pH of buffer with 0.11 M CH_3COONa and 0.09 M CH_3COOH. Ka = 1.85×10^{-5}.

pKa = 4.73. pH = 4.73 + log(0.11/0.09) = 4.82.

Quick Check 9.5 (book p. 199)

a. Ksp(AgCl) = 1.77×10^{-10}. Find solubility of AgCl in 0.10 M NaCl.

With common Cl^- = 0.10 M, solubility s = [Ag^+] = Ksp/[Cl^-] = $1.77\times10^{-10}/0.10$ = 1.77×10^{-9} mol dm^{-3}.

b. Will $CaSO_4$ precipitate when 10^{-3} M Ca^{2+} and 10^{-3} M SO_4^{2-} are mixed?

Ionic product = $(10^{-3})(10^{-3}) = 10^{-6}$. Since this is less than Ksp($CaSO_4$) (~2.4×10^{-5}), no precipitate forms.

c. Why K_2CO_3 solution is basic; give hydrolysis equation.

CO_3^{2-} is the conjugate base of the weak acid HCO_3^-, so it hydrolyses: $CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$, producing OH^- and making the solution alkaline.

Quick Check 9.6 (book p. 204)

a. Differentiate end point and equivalence point.

The equivalence point is the stoichiometric point where acid and base have reacted completely in mole ratio. The end point is the observed indicator colour change, which should be close to the equivalence point.

b. How an indicator changes colour in acidic/basic solution.

An indicator is a weak acid/base ($HIn \rightleftharpoons H^+ + In^-$) whose two forms have different colours. In acid the equilibrium shifts to HIn (acid colour); in base it shifts to In^- (base colour).

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c. Suitable indicator for weak acid–strong base titration.

Phenolphthalein (equivalence point is basic, within its colour-change range).

d. Suitable indicator for weak acid–weak base titration.

No common indicator gives a sharp end point because the pH change near equivalence is too gradual; a pH meter is preferred.

Textbook checks flagged in this chapter

• p. 205, Q.2(a): two sub-items are printed as “iii)”; both are answered.

• p. 205, Q.2(g): Ksp(CuS) is printed without an equals sign; the value 8×10^{-34} is used.

• p. 204, MCQ IV: both phenolphthalein and bromothymol blue are suitable for strong acid–strong base titrations.

• p. 205, MCQ VII: the equivalence point (stoichiometric) is conflated with the indicator end point.

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